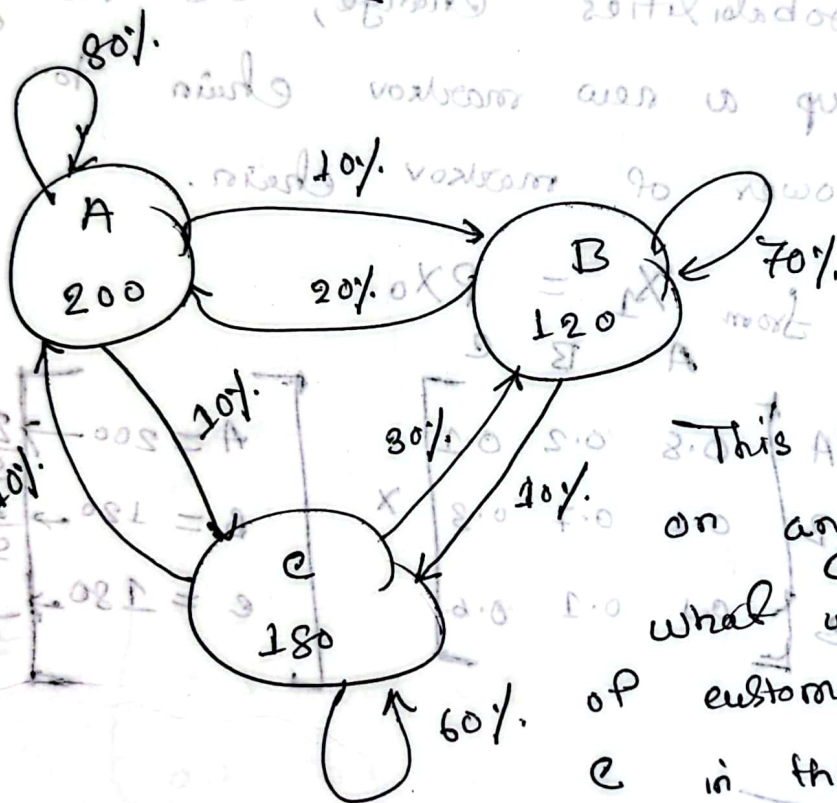


Markov chain.

$$\begin{bmatrix} \text{next state} \\ \text{future state.} \end{bmatrix} = \begin{bmatrix} \text{matrix of} \\ \text{transition} \\ \text{probabilities} \end{bmatrix} \times \begin{bmatrix} \text{current state} \end{bmatrix}$$



This is the scenario on any given week. what will be the number of customers on shop A, B, C in the following week?

If the situation remains the same (meaning if the transition probabilities remains the same), what will be the number of customers in

grocery shop in A, B, C after that. the week after that, the week after that?

If the probabilities remain the same, we can predict what will be the customer base for each of the stores & eventually they will converge to particular numbers. That's the power of Markov chain.

If the probabilities change, we can again have to set up a new Markov chain to see the predictive power of Markov chain.

$$\text{from } X_1 = P X_0$$

$$\begin{bmatrix} A_1 \\ B_1 \\ C_1 \end{bmatrix} = \begin{matrix} A \\ B \\ C \end{matrix} \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \times \begin{bmatrix} A = 200 \\ B = 120 \\ C = 180 \end{bmatrix}$$

$\frac{200}{500} = 0.40$
 $\frac{120}{500} = 0.24$
 $\frac{180}{500} = 0.36$

$$\begin{bmatrix} 0.8 \times 0.4 + 0.2 \times 0.24 + 0.1 \times 0.36 \\ 0.1 \times 0.4 + 0.7 \times 0.24 + 0.3 \times 0.36 \\ 0.1 \times 0.4 + 0.1 \times 0.24 + 0.6 \times 0.36 \end{bmatrix}$$

$$= \begin{bmatrix} 202 \\ 158 \\ 140 \end{bmatrix} = \begin{bmatrix} 0.404 \\ 0.316 \\ 0.280 \end{bmatrix} = \begin{bmatrix} 500 \times 0.404 \\ 500 \times 0.316 \\ 500 \times 0.280 \end{bmatrix}$$

$$X_1 = P X_0$$

$$X_2 = P X_1$$

$$X_3 = P X_2$$

...

$$X_1 = \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} X_0 = \begin{bmatrix} 0.40 \\ 0.24 \\ 0.36 \end{bmatrix}$$

$$= \begin{bmatrix} 0.404 \\ 0.316 \\ 0.280 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} X_1 = \begin{bmatrix} 0.404 \\ 0.316 \\ 0.280 \end{bmatrix}$$

$$= \begin{bmatrix} 0.414 \\ 0.346 \\ 0.240 \end{bmatrix}$$

$$X_3 = P X_2$$

~~P~~

$$X_1 = PX_0$$

$$X_2 = PX_1$$

$$= P \times PX_0$$

$$X_3 = P X_2 = P^2 X_0$$

0.14.0	5.0	8.0
1.5.0	2.0	7.0
2.8.0	1.0	1.0

$$= 1X$$

$$= \cancel{P X_0}$$

$$= P \times P^2 X_0$$

$$= P^3 X_0$$

0.14.0
2.18.0
0.85.0

$$=$$

$$X_4 = P X_3 = P \times P^3 X_0 = P^4 X_0$$

0.14.0	5.0	8.0
2.18.0	2.0	7.0
0.85.0	1.0	1.0

$$= 5X$$

$$\boxed{X_n = P^n X_0}$$

0.14.0
2.18.0
0.85.0

$$=$$

$$\begin{aligned}
 X_1 &= P X_0 \\
 &= \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \times \begin{bmatrix} 0.40 \\ 0.24 \\ 0.36 \end{bmatrix} \\
 &= \begin{bmatrix} 0.404 \\ 0.346 \\ 0.280 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 X_2 &= \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \times \begin{bmatrix} 0.404 \\ 0.346 \\ 0.280 \end{bmatrix} \\
 &= \begin{bmatrix} 0.414 \\ 0.346 \\ 0.240 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 X_3 &= \begin{bmatrix} 0.8 & 0.2 & 0.4 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \times \begin{bmatrix} 0.414 \\ 0.346 \\ 0.240 \end{bmatrix} \\
 &= \begin{bmatrix} 0.424 \\ 0.356 \\ 0.220 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 X_4 &= \begin{bmatrix} 0.8 & 0.2 & 0.4 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \times \begin{bmatrix} 0.424 \\ 0.356 \\ 0.220 \end{bmatrix} \\
 &= \begin{bmatrix} 0.432 \\ 0.358 \\ 0.210 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 X_5 &= \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \times \begin{bmatrix} 0.432 \\ 0.358 \\ 0.210 \end{bmatrix} \\
 &= \begin{bmatrix} 0.438 \\ 0.357 \\ 0.205 \end{bmatrix}
 \end{aligned}$$

$\bar{X} = \text{average}$

$$X_6 = \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \times \begin{bmatrix} 0.438 \\ 0.357 \\ 0.205 \end{bmatrix} = X$$

$$= \begin{bmatrix} 0.442 \\ 0.355 \\ 0.203 \end{bmatrix}$$

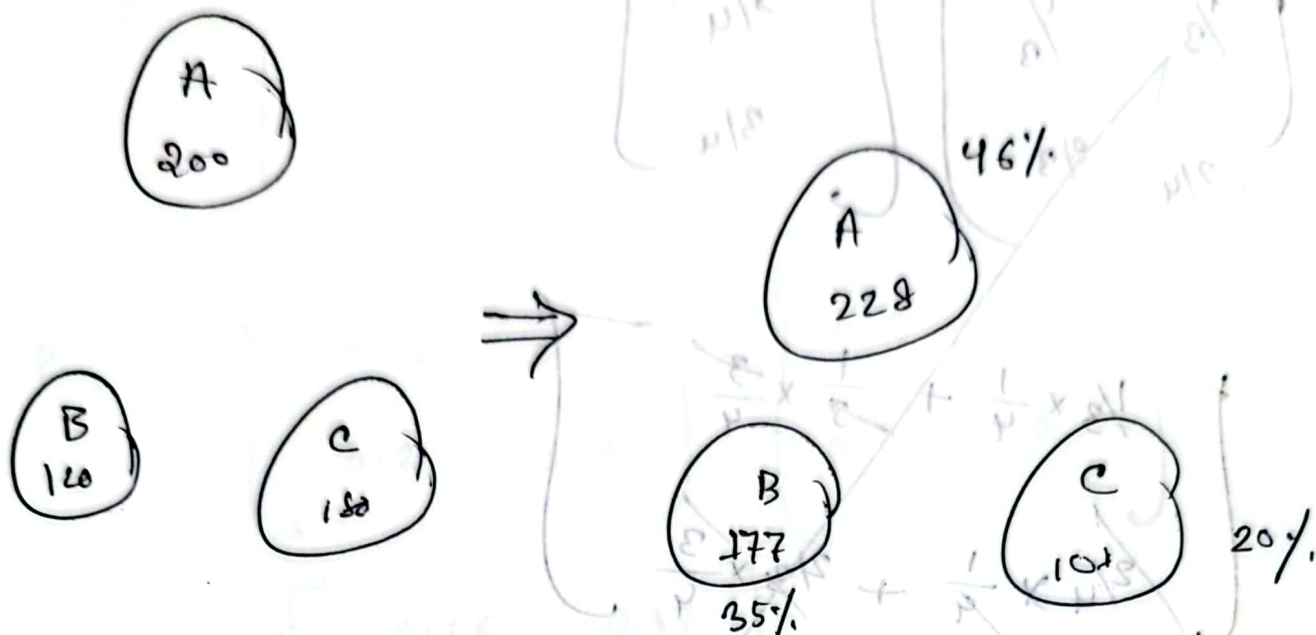
$$X_7 = \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \times \begin{bmatrix} 0.442 \\ 0.355 \\ 0.203 \end{bmatrix} = X$$

$$= \begin{bmatrix} 0.445 \\ 0.354 \\ 0.202 \end{bmatrix}$$

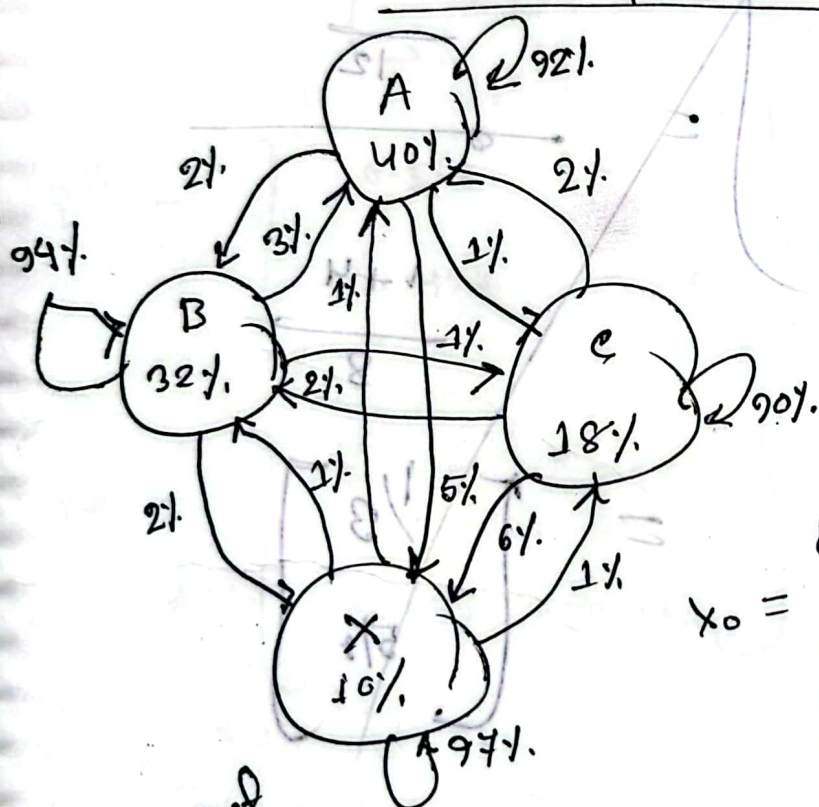
$$X_8 = \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \times \begin{bmatrix} 0.445 \\ 0.354 \\ 0.202 \end{bmatrix} = X$$

$$= \begin{bmatrix} 0.445 \\ 0.354 \\ 0.202 \end{bmatrix}$$

Stable ~~dist~~ distribution
matrix = $\bar{X}$



Market Penetration



$P =$

	A	B	C	X	
A	0.92	0.03	0.02	0.01	A
B	0.02	0.94	0.02	0.01	B
C	0.01	0.01	0.90	0.01	C
X	0.05	0.02	0.06	0.97	X

$X_0 =$

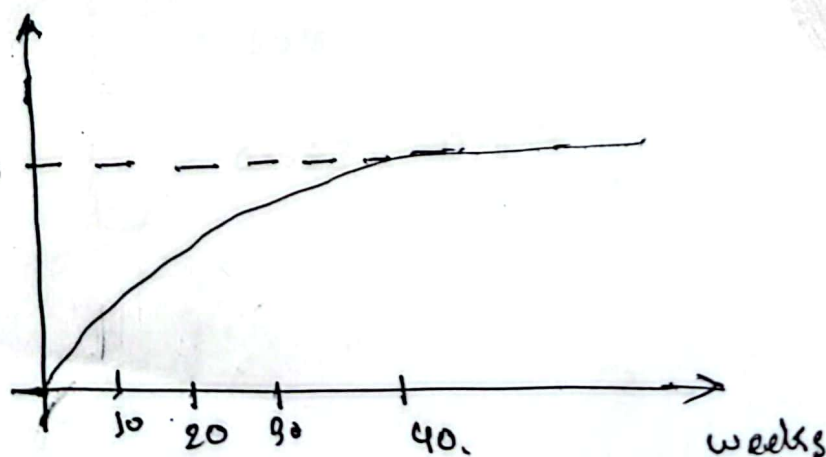
0.40
0.32
0.18
0.10

$\bar{X} =$

0.161
0.179
0.091
0.569

Market Share

56.9%



$$P = \begin{matrix} & \begin{matrix} A & B \end{matrix} \\ \begin{matrix} A \\ B \end{matrix} & \begin{bmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{bmatrix} \end{matrix}$$

$$P^2 = \begin{bmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{bmatrix} \lambda \begin{bmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{bmatrix}$$

$$= \begin{bmatrix} 0.66 & 0.17 \\ 0.34 & 0.83 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.562 & 0.249 \\ 0.438 & 0.751 \end{bmatrix}, P^4 = \begin{bmatrix} 0.493 & 0.253 \\ 0.507 & 0.747 \end{bmatrix}$$

$$P^5 = \begin{bmatrix} 0.445 & 0.277 \\ 0.555 & 0.723 \end{bmatrix}$$

$$\bar{P} = \begin{bmatrix} 0.333 & 0.333 \\ 0.667 & 0.667 \end{bmatrix} A = \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}$$

$$X_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$X_0 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

$$X_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\bar{P} X_0 = \bar{X}$$

Probability matrix in Markov chain has two properties.

(i)

they must be stochastic.

(ii)

they must be regular.

from

$$P = \begin{bmatrix} A & B \\ 0.25 & 0.50 \\ 0.75 & 0.50 \end{bmatrix}$$

A B



add to 1.



add to 1.

otherwise not stochastic.

you can't do markov chain if the matrix is not stochastic.

Stochastic, $P = \begin{bmatrix} 0.25 & 0.50 \\ 0.75 & 0.50 \end{bmatrix}$ $\bar{X} = \infty$

A stochastic matrix is also a regular matrix.

P^n ($n > 1$) has only non-zero positive entries.

$$P = \begin{bmatrix} 0.25 & 0.50 \\ 0.75 & 0.50 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0.5 \\ 0 & 0.5 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.438 & 0.375 \\ 0.562 & 0.625 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 1 & 0.75 \\ 0 & 0.25 \end{bmatrix}$$

regular

non-regular

$$P = \begin{bmatrix} 0.25 & 0.50 \\ 0.75 & 0.50 \end{bmatrix}$$

chain

the transition matrix in Markov must be stochastic & regular

$$P = \begin{bmatrix} 0.25 & 0.50 \\ 0.75 & 0.50 \end{bmatrix}$$

$$P = \begin{bmatrix} 0.25 & 0.50 \\ 0.75 & 0.50 \end{bmatrix}$$

stable = $\bar{X} = \infty$

If P is a regular matrix,
there will be a P^n where

$$P^n x_0 = \bar{x}$$

Stable distribution matrix.

or

where

$$P^{n+1} = P^n$$

then

$P, P^2, P^3, P^4, P^5, \dots$ is a regular Markov chain.

$$x_0 = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}$$

$$P = \begin{bmatrix} 0.4 & 0.3 \\ 0.6 & 0.7 \end{bmatrix}$$

$$P^4 x_0 = \begin{bmatrix} 0.333 & 0.333 \\ 0.667 & 0.667 \end{bmatrix} \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 0.334 & 0.333 \\ 0.666 & 0.667 \end{bmatrix}$$

$$= \begin{bmatrix} 0.333 \\ 0.667 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.334 & 0.333 \\ 0.666 & 0.667 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 0.333 & 0.333 \\ 0.667 & 0.667 \end{bmatrix}$$

$= \bar{P} =$ Stable transition Probability matrix.

let

$$P = \begin{bmatrix} 0.5 & 0.25 \\ 0.5 & 0.75 \end{bmatrix} \text{ and}$$

$\bar{P}$ is the stable matrix = $\begin{bmatrix} 0.333 & 0.333 \\ 0.667 & 0.667 \end{bmatrix}$

$$X_0 = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}$$

$\bar{X}$ is the stable distribution matrix

$$\begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 1/3 \\ 2/3 \end{bmatrix}$$

Then, $P\bar{X} = \bar{X}$

So that, $\bar{P}X_0 = \bar{X}$

$$\Rightarrow \begin{bmatrix} 0.5 & 0.25 \\ 0.5 & 0.75 \end{bmatrix} \begin{bmatrix} 1/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 0.333 & 0.333 \\ 0.667 & 0.667 \end{bmatrix} \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} = \begin{bmatrix} 0.333 \\ 0.667 \end{bmatrix}$$

$$= \begin{bmatrix} 0.333 \\ 0.667 \end{bmatrix}$$

Q. If $P = \begin{bmatrix} 0.5 & 0.25 \\ 0.5 & 0.75 \end{bmatrix}$ from

So, and is $\bar{P} = 3$ $\bar{X} = 3$

let, $\bar{X} = \begin{bmatrix} A \\ B \end{bmatrix}$

$P\bar{X} = \bar{X}$

$\Rightarrow \begin{bmatrix} 0.5 & 0.25 \\ 0.5 & 0.75 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} A \\ B \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 0.5A + 0.25B \\ 0.5A + 0.75B \end{bmatrix} = \begin{bmatrix} A \\ B \end{bmatrix}$

$\Rightarrow 0.5A + 0.25B = A$

$0.5A + 0.75B = B$

and $A + B = 1$

(i)

(ii)

(iii)

$$A = 1/3$$

$$B = 2/3$$

$$\bar{X} = \begin{bmatrix} 1/3 \\ 2/3 \end{bmatrix}$$

$$\bar{P} = \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}$$

Q

$$P = \begin{bmatrix} 0.9 & 0.5 \\ 0.1 & 0.5 \end{bmatrix}, \quad \bar{P} = ? \quad \& \quad \bar{X} = ?$$

$$\text{let, } \bar{X} = \begin{bmatrix} A \\ B \end{bmatrix}$$

$$P\bar{X} = \bar{X}$$

$$\Rightarrow \begin{bmatrix} 0.9 & 0.5 \\ 0.1 & 0.5 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} A \\ B \end{bmatrix}$$

$$\therefore A = 5/6 \quad \& \quad B = 1/6$$

$$\bar{X} = \begin{bmatrix} 5/6 \\ 1/6 \end{bmatrix}$$

$$\bar{P} = \begin{bmatrix} 5/6 & 5/6 \\ 1/6 & 1/6 \end{bmatrix}$$

from $\rightarrow$

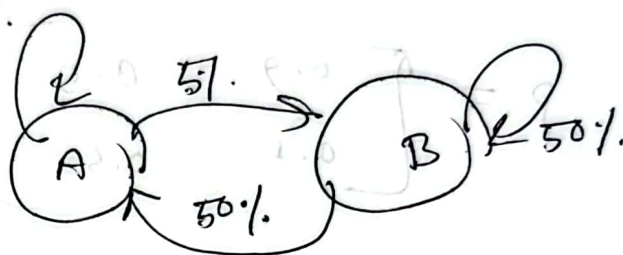
Q

$$P = \begin{bmatrix} 0.95 & 0.5 \\ 0.05 & 0.5 \end{bmatrix} \begin{matrix} A \\ B \end{matrix} \downarrow \varnothing.$$

Find $\bar{P} = ?$ & $\bar{X} = ?$

Let, $\bar{X} = \begin{bmatrix} A \\ B \end{bmatrix}$ 95%

$$P\bar{X} = \bar{X}$$



$$\Rightarrow \begin{bmatrix} 0.95 & 0.5 \\ 0.05 & 0.5 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} A \\ B \end{bmatrix}$$

$$A + B = 1.$$

$$\therefore A = \frac{10}{11}, B = \frac{1}{11}$$

$$\bar{P} = \begin{bmatrix} \frac{10}{11} & \frac{10}{11} \\ \frac{1}{11} & \frac{1}{11} \end{bmatrix} \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}$$

$$\frac{10}{11} \times 0.4 + 0.6 \times \frac{10}{11}$$

$$= \frac{10 \times 0.4 + 0.6 \times 10}{11}$$

Q $P = \begin{bmatrix} & A & B & C \\ \begin{bmatrix} 0.8 & 0.2 & 0.3 \\ 0.1 & 0.7 & 0.1 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} & \begin{bmatrix} A \\ B \\ C \end{bmatrix} \end{bmatrix}$

Find $\bar{x} = ?$ & $\bar{p} = ?$

Soln: $\bar{x} = \begin{bmatrix} A \\ B \\ C \end{bmatrix}$

$P\bar{x} = \bar{x}$

$$\Rightarrow \begin{bmatrix} 0.8 & 0.2 & 0.3 \\ 0.1 & 0.7 & 0.1 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} A \\ B \\ C \end{bmatrix}$$

$\Rightarrow 0.8A + 0.2B + 0.3C = A$

$0.1A + 0.7B + 0.1C = B$

$0.1A + 0.1B + 0.6C = C$

$A + B + C = 1$

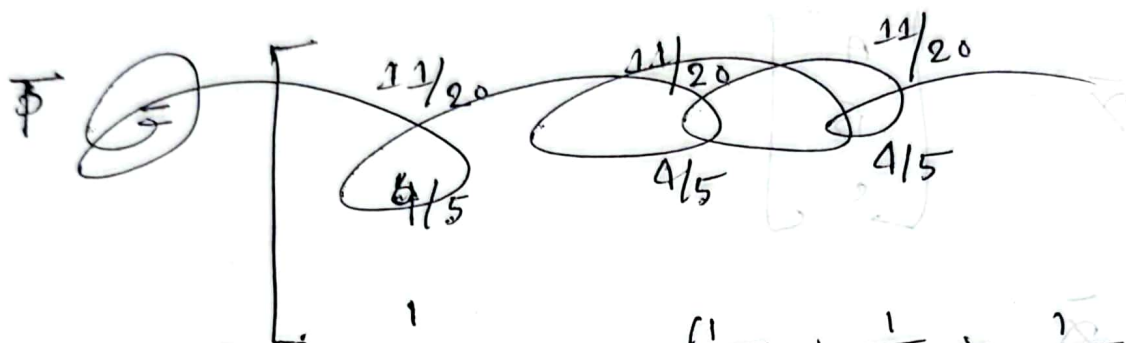
~~$B = \frac{1}{4}, C = \frac{4}{5}, A = \frac{11}{20}$~~

$$\bar{X} = \begin{bmatrix} \frac{11}{20} \\ \frac{1}{4} \\ \frac{1}{5} \end{bmatrix}$$

$$A = \frac{11}{20}$$

$$B = \frac{1}{4}$$

$$C = \frac{1}{5}$$



$$\begin{aligned} & \frac{11}{20} + \frac{1}{4} + \frac{1}{5} \\ &= \frac{11 + 5 + 4}{20} = \frac{20}{20} = 1. \end{aligned}$$

$$\bar{P} = \begin{bmatrix} \frac{11}{20} & \frac{11}{20} & \frac{11}{20} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \end{bmatrix}$$

Q charity contribution $\Rightarrow$ of those who contribute 40% will not contribute next time. of those who don't contribute 10% will contribute next time. what is the final distribution matrix = ?

Solⁿ:
$$P = \begin{matrix} & \begin{matrix} N & C \end{matrix} \\ \begin{matrix} N \\ C \end{matrix} & \begin{bmatrix} 0.9 & 0.4 \\ 0.1 & 0.6 \end{bmatrix} \end{matrix}$$

Now find $\bar{x} = ?$

Q Grocery Stores $\Rightarrow$ Two stores A & B.

30% of customers shopping at A will switch to B every month. 20% of customers shopping at B will switch to A every month. what is the stable distribution matrix = ?

Solⁿ:
$$P = \begin{matrix} & \begin{matrix} A & B \end{matrix} \\ \begin{matrix} A \\ B \end{matrix} & \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix} \end{matrix}$$

$\lambda = ?$

Hidden Markov Model

Example

$Q = q_1, q_2, \dots, q_n$ a set of states

$A = a_{11}, a_{12}, \dots, a_{nn}$

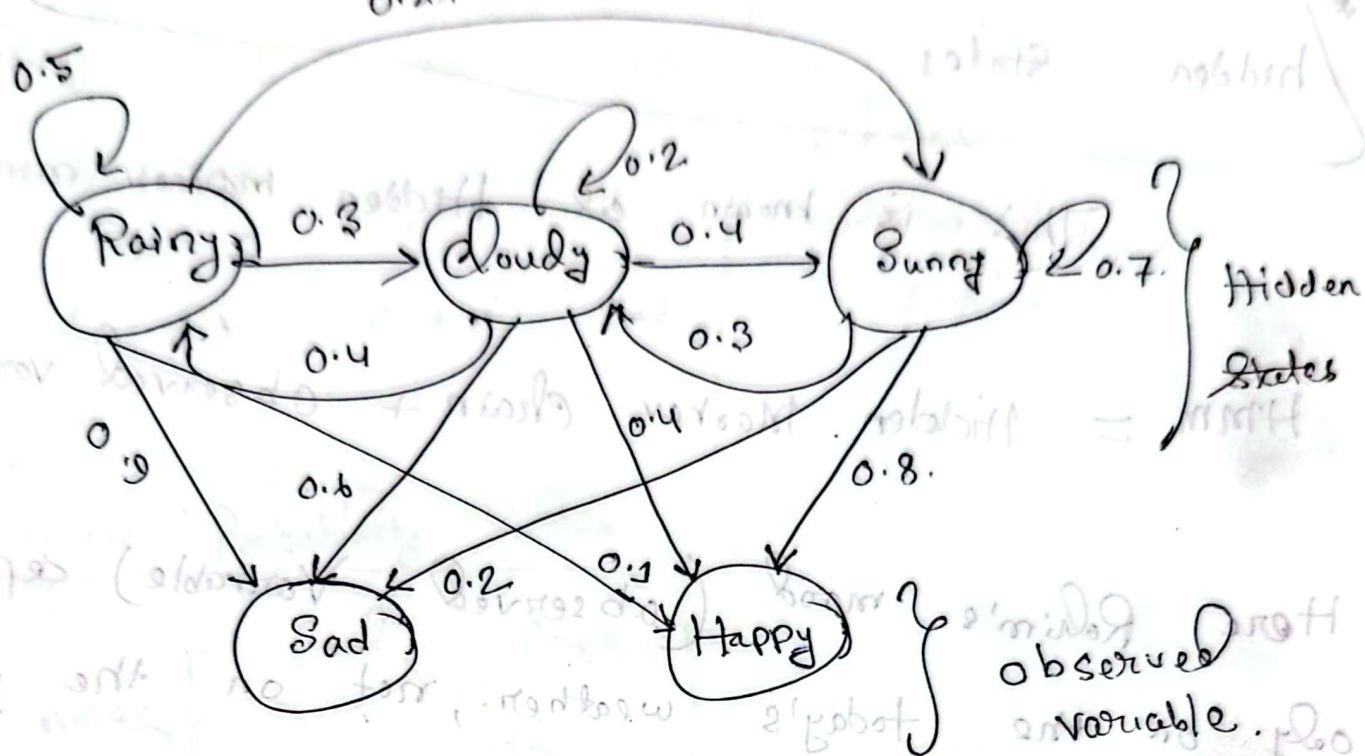
$\pi = \pi_1, \pi_2, \dots, \pi_n$

$$\sum_{j=1}^n a_{ij} = 1 \quad \forall i$$

	A	C	G	T
A	0.18	0.27	0.43	0.12
C	0.17	0.57	0.27	0.19
G	0.16	0.34	0.38	0.12
T	0.08	0.31	0.38	0.18

$(C G A T) = 3$

Hidden Markov Model



Suppose, your friend Rahim's city weather is depicted on any given day in the above diagram. And the mood of your friend depends on the current day, not on the previous day. Somehow, it is possible for you to contact your friend to know about his mood but it is not possible for you to know the weather. So the Markov chain of weather is hidden here. We can only observe the variable whether Mr. Rahim is Sad or Happy.

* Observed variables are dependent on the hidden states.

This is known as Hidden Markov model

HMM = Hidden Markov chain + observed variables.

Here Rahim's mood (Observed Variable) depends only on the today's weather, not on the previous day's mood.

Transition Probability Matrix:

	Rainy	Cloudy	Sunny
Rainy	0.5	0.3	0.2
Cloudy	0.4	0.2	0.4
Sunny	0.0	0.3	0.7

From

To \ From	Rainy	cloudy	sunny
Rainy	0.5	0.4	0.0
cloudy	0.3	0.2	0.3
sunny	0.2	0.4	0.7

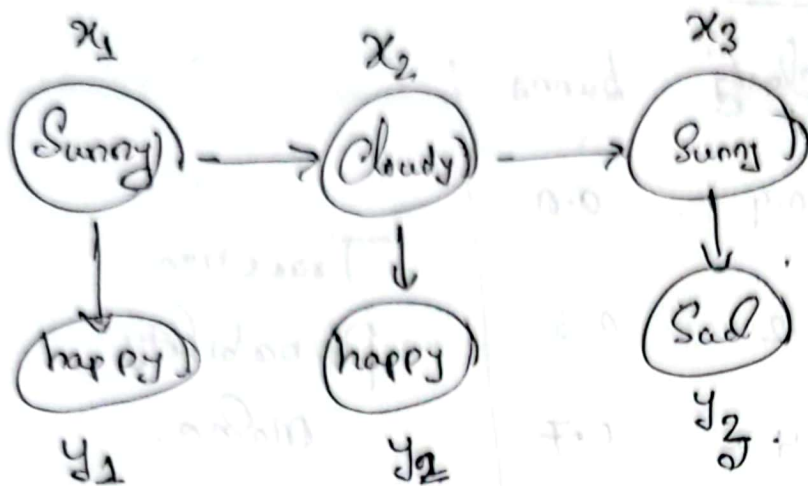
Transaction Probability Matrix.

Emission Probability Matrix:

	Sad	Happy
Rainy		
cloudy		
Sunny		

from

	Rainy	cloudy	Sunny
Sad	0.9	0.6	0.2
Happy	0.1	0.4	0.8



What is the probability of this scenario to happen?

or

What is the joint probability of observed mood seq & weather seq?

$$P(Y = \frac{Y_1}{\text{happy}}, \frac{Y_2}{\text{happy}}, \frac{Y_3}{\text{Sad}} \mid X = \frac{X_1}{\text{Sunny}}, \frac{X_2}{\text{cloudy}}, \frac{X_3}{\text{Sunny}})$$

$$P(X_1 = \text{Sunny}) \cdot P(Y_1 = \text{happy} \mid X_1 = \text{Sunny})$$

~~X~~

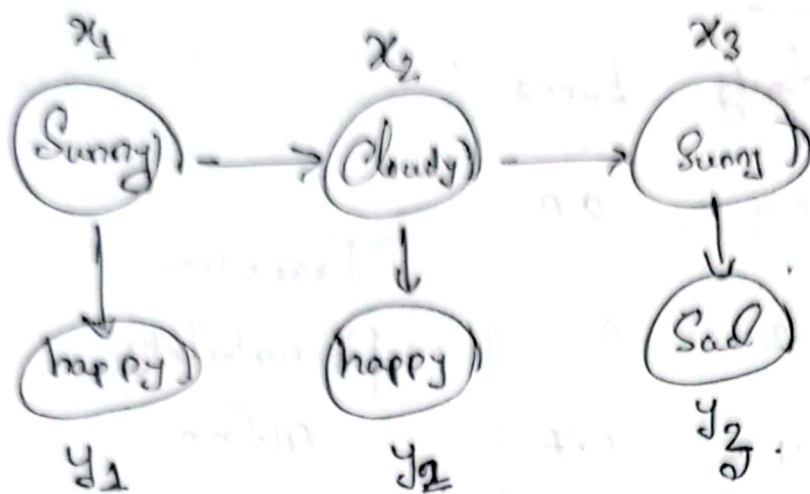
$$P(X_2 = \text{cloudy} \mid X_1 = \text{Sunny}) \cdot P(Y_2 = \text{happy} \mid X_2 = \text{cloudy})$$

$$P(X_3 = \text{Sunny} \mid X_2 = \text{cloudy}) \cdot P(Y_3 = \text{Sad} \mid X_3 = \text{Sunny})$$

Now what is the most likely weather seq. for the observed mood seq? (Happy, Happy, Sad)?

$$P(Y = \text{happy, happy, sad} \mid X = \text{Sunny, Sunny, cloudy}) = 0.04105$$

Ans: $0.509 \times 0.8 \times 0.3 \times 0.4 \times 0.4 \times 0.2 = 0.00391$



What is the probability of this scenario to happen?

or

What is the joint probability of observed mood seqⁿ & weather seqⁿ?

$$P\left(Y = \frac{Y_1}{\text{happy}}, \frac{Y_2}{\text{happy}}, \frac{Y_3}{\text{Sad}} \mid X = \frac{X_1}{\text{Sunny}}, \frac{X_2}{\text{Cloudy}}, \frac{X_3}{\text{Sunny}}\right)$$

$$P(X_1 = \text{Sunny}) \cdot P(Y_1 = \text{happy} \mid X_1 = \text{Sunny})$$

Ans: $0.509 \times 0.8 \times 0.3 \times 0.4$
 $\times 0.4 \times 0.2$
 $= 0.00391$

$$P(X_2 = \text{Cloudy} \mid X_1 = \text{Sunny}) \cdot P(Y_2 = \text{happy} \mid X_2 = \text{Cloudy})$$

$$P(X_3 = \text{Sunny} \mid X_2 = \text{Cloudy}) \cdot P(Y_3 = \text{Sad} \mid X_3 = \text{Sunny})$$

Now what is the most likely weather seqⁿ for the observed mood seqⁿ? (Happy, Happy, Sad)?

Ans: $P(Y = \text{happy, happy, sad} \mid X = \text{Sunny, Sunny, Cloudy}) = 0.04105$

If the hidden states are represented by x and observed variables are by y .

$$\arg \max_{x = x_1, x_2, \dots, x_n} P(x = x_1, x_2, \dots, x_n | y = y_1, y_2, \dots, y_n)$$

meaning find the particular seqⁿ of x for

which x given y is maximum.

Baye's theorem:

$$P(x|y) = \frac{P(y|x) P(x)}{P(y)}$$

$$P(x|y) = \frac{P(y|x) P(x)}{P(y)}$$

$$P(x = x_1, x_2, \dots, x_n | y = y_1, y_2, \dots, y_n) = \arg \max_{x = x_1, x_2, \dots, x_n} \frac{P(y|x) P(x)}{P(y)}$$

$$P(y|x) = P(y_1|x_1) * P(y_2|x_2) * P(y_3|x_3) * \dots * P(y_n|x_n).$$

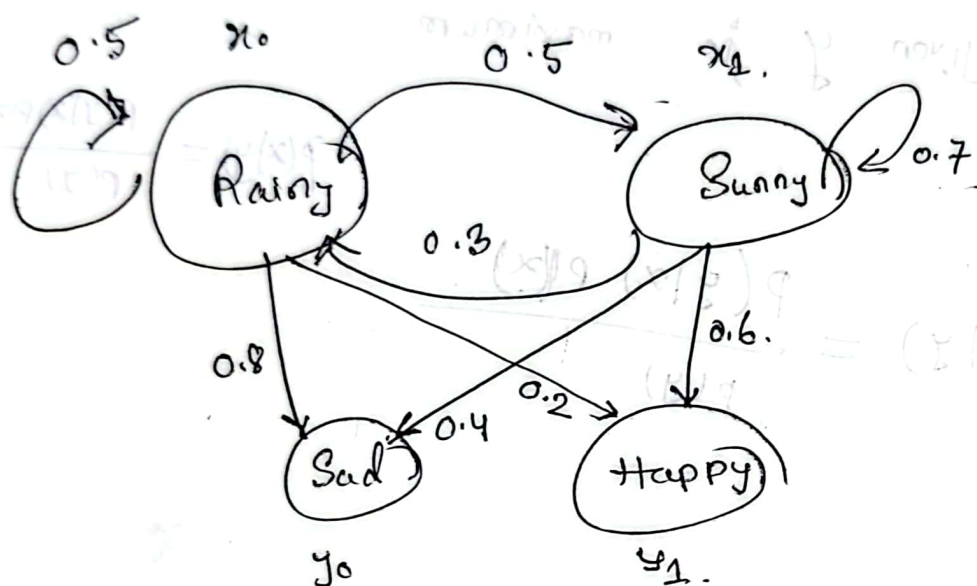
$$P(y|x) = \prod_{i=1}^n P(y_i|x_i)$$

$$P(x) = \prod_{i=1}^n P(x_i|x_{i-1}).$$

We can omit $P(y)$ for practical purpose as it is same for all cases.

$$P(x=x_1, x_2, \dots, x_n | y=y_1, y_2, \dots, y_n) = \arg \max_{\text{max}} \prod_{i=1}^n p(y_i | x_i) p(x_i | x_{i-1})$$

Forward algorithm

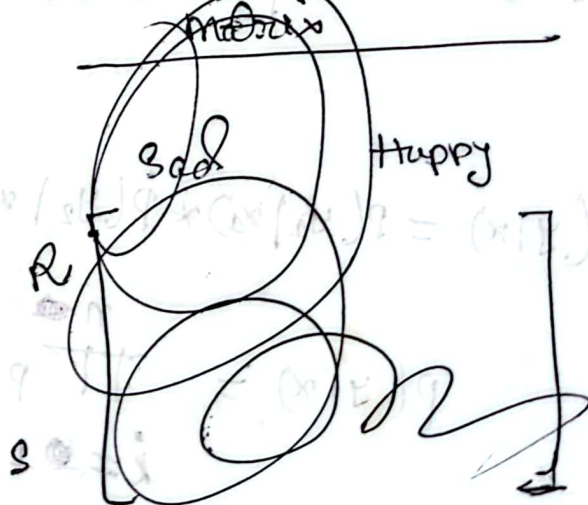


Transition Probability matrix

from

$$R \begin{bmatrix} 0.5 & 0.3 \\ 0.5 & 0.7 \end{bmatrix}$$

Emission Probability matrix



Emission Probability Matrix:

	Rainy (R)	Sunny (S)
Sad	0.8	0.4
Happy	0.2	0.6

Calculate the stable Distribution matrix:

$$\begin{bmatrix} 0.5 & 0.3 \\ 0.5 & 0.7 \end{bmatrix} \begin{bmatrix} R \\ S \end{bmatrix} = \begin{bmatrix} R \\ S \end{bmatrix}$$

$$0.5R + 0.3S = R$$

$$0.5R + 0.7S = S$$

$$R + S = 1$$

Solving them,

$$\pi = \begin{bmatrix} 0.375 \\ 0.625 \end{bmatrix} \begin{matrix} \rightarrow P(\text{Rainy}) \\ \rightarrow P(\text{Sunny}) \end{matrix}$$

$$\bar{x} = P\bar{x}$$

$$\Rightarrow \begin{bmatrix} R \\ S \end{bmatrix} = \begin{bmatrix} R \\ S \end{bmatrix}$$

$$\begin{bmatrix} R \\ S \end{bmatrix}$$

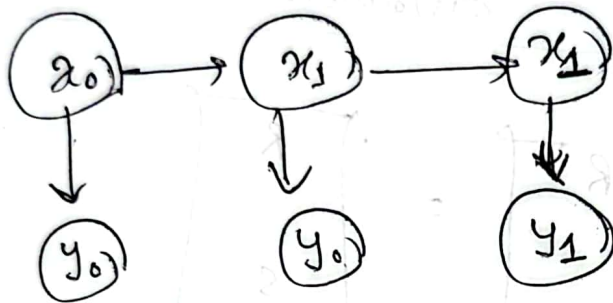
$$R + S = 1$$

Q What is the probability of this observed seq
given the parameters of our HMM?

$y = \text{sad, sad, Happy.}$

~~Q~~ $p(y = y_0, y_0, y_1) = ?$

if we know the scenario as.



$$p(x_0) p(y_0|x_0) * p(x_1|x_0) p(y_0|x_1) * p(x_2|x_1) p(y_1|x_2)$$

$2^3 = 8$ possible permutations:

$$\begin{aligned}
 & p(x_0) p(y_0|x_0) p(x_1|x_0) p(y_0|x_1) p(x_2|x_1) p(y_1|x_2), \\
 & p(x_0) p(y_0|x_0) p(x_1|x_0) p(y_1|x_1) p(x_2|x_1) p(y_1|x_2), \\
 & p(x_0) p(y_0|x_0) p(x_1|x_0) p(y_0|x_1) p(x_0|x_1) p(y_1|x_0), \\
 & p(x_0) p(y_0|x_0) p(x_1|x_0) p(y_0|x_1) p(x_1|x_1) p(y_1|x_1), \\
 & p(x_1) p(x_0|x_1) p(x_0|x_1) p(y_0|x_0) p(x_0|x_0) p(y_1|x_0), \\
 & p(x_1) p(y_0|x_1) p(x_0|x_1) p(y_0|x_0) p(x_1|x_0) p(y_1|x_1), \\
 & p(x_1) p(y_0|x_1) p(x_0|x_1) p(y_0|x_1) p(x_0|x_1) p(y_1|x_0), \\
 & p(x_1) p(y_0|x_1) p(x_1|x_1) p(y_0|x_1) p(x_1|x_1) p(y_1|x_1).
 \end{aligned}$$

number of
multiplications
required = $2 + N^T$

Hidden Markov model

allows us to talk about both observed events and hidden events that we think of as causal factors in our probabilistic model. An HMM is specified by the following components:

$X = x_1, x_2, \dots, x_n$ a set of n states.

$A = a_{11}, a_{12}, \dots, a_{nn}$, a transition probability matrix

where a_{ij} represents the prob. of moving from state i to

state j .

$$\sum_{j=1}^n a_{ij} = 1 \quad \forall i$$

$Y = y_1, y_2, \dots, y_m$, a sequence of m observations, each one drawn from a vocabulary

$$V = v_1, v_2, \dots, v_V$$

$B = P(y_i | x_i)$ a sequence of observation likelihoods, also called emission probabilities, each expressing the prob. of an observation y_i being generated from state x_i

$\pi = \pi_1, \pi_2 \dots \pi_n$, an initial probability distribution over states $\dots \pi_i$ is the prob. that the markov chain will start in state i .

$$\sum_{i=1}^n \pi_i = 1$$

Forward algorithm.

Function Forward(observations of length n , state-graph of length n)
returns forward-prob.

create a probability matrix,

forward $[n, m]$

↙ ni. of states
↓
no. of observations

for each state s from 1 to n do:

$$\text{forward}[s, 1] = \pi_s * b_s(o_1)$$

Initialization Step:

for each time step t from 2 to m do;

for each state s from 1 to n do:

$$\text{forward}[s, t] = \sum_{s'=1}^n \text{forward}[s', t-1] * a_{s', s} * b_s(a_t).$$

recursion step.

$$\text{forward prob} \rightarrow \sum_{s=1}^n \text{forward}[s, m]$$

termination step

Ans:

m
neighbours.



A & B are connected by a single node.

Input:

Output:

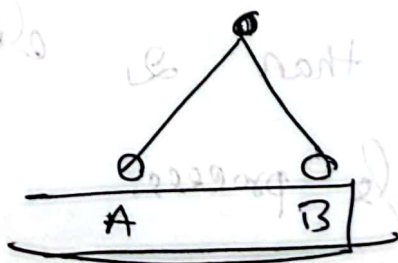
unrooted tree with branch lengths.

repeatedly joins pairs of leaves / subtrees by rules of numerical optimization.

It shrinks the distance matrix by considering two neighbours as one node.

It shrinks the distance matrix by considering two neighbours as one node.

It shrinks the distance matrix by considering two neighbours as one node.



$$\delta_i = \sum_{j=1}^n d_{ij}$$

For all pair of ~~formulated~~ nodes,

$$m(i, j) = d(i, j) - \frac{\delta_i + \delta_j}{n-2}$$

n = no. of nodes.

2/ Join i & j to a new node u . The branch length from u to i & j are:

$$b(u, i) = \frac{d(i, j)}{2} + \frac{\delta_i - \delta_j}{2n-4}$$

$$b(u, j) = d(i, j) - b(u, i)$$

3/ Now update the distance matrix:

the distance from u to other nodes

k ,

$$d(u, k) = \frac{d(i, k) + d(j, k) - d(i, j)}{2}$$

If there are more than 2 clusters left then repeat the whole process.

	A	B	C	D
A	0	3	4	5
B	3	0	5	4
C	4	5	0	7
D	5	4	7	0

$$\gamma_A = 3 + 4 + 5 = 12$$

$$\gamma_B = 3 + 5 + 4 = 12$$

$$\gamma_C = 4 + 5 + 7 = 16$$

$$\gamma_D = 5 + 4 + 7 = 16$$

$$m(A, B) = d(A, B) - \frac{\gamma_A + \gamma_B}{n-2}$$

$$= 3 - \frac{12+12}{4-2}$$

$$= 3 - 12 = -9$$

$$m(A, C) = d(A, C) - \frac{\gamma_A + \gamma_C}{n-2}$$

$$= 4 - \frac{12+16}{2}$$

$$= 4 - 14 = -10$$

$$m(A, D) = d(A, D) - \frac{\gamma_A + \gamma_D}{n-2}$$

$$= 5 - 14$$

$$= -9$$

$$m(B, C) = d(B, C) - \frac{\gamma_B + \gamma_C}{n-2}$$

$$= 5 - \frac{12+16}{2}$$

$$= -9$$

$$m(B, D) = 4 - 14$$

$$= -10$$

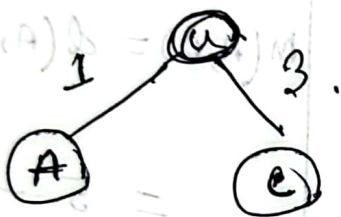
$$m(C, D) = 7 - \frac{16+16}{2}$$

$$= -9$$

	A	B	C	D
A	-	-9	-10	-9
B		-	-9	-10
C			-	-9
D				-

$m(i, j)$
M-table:

$m(A, e)$ & $m(B, d)$



Branch length from u to A & e :

$$b(u, A) = \frac{d(A, e)}{2} + \frac{\gamma_A - \gamma_e}{2n - 4}$$

$$= \frac{4}{2} + \frac{12 - 16}{2 \times 4 - 4}$$

$$= 2 + \frac{-4}{4}$$

$$= 1$$

$$b(u, e) = d(A, e) - b(u, A)$$

$$= 4 - 1$$

$$= 3$$

~~$$b(u, e) = \frac{d(A, e)}{2} + \frac{\gamma_A - \gamma_e}{2n - 4}$$

$$= \frac{4}{2} + \frac{12 - 16}{2 \times 4 - 4}$$

$$= 2 + \frac{-4}{4}$$

$$= 1$$~~

update the distance matrix

$d(i, j)$:

	A _c	B	D
A _c	0	2	4
B	2	0	4
D	4	4	0

$u = A_c$

$$d(u, w) = \frac{d(i, k) + d(j, w) - d(i, j)}{2}$$

$$d(A_c, B) = \frac{d(A, B) + d(A_c, B) - d(A, A_c)}{2}$$

$$= \frac{3 + 5 - 4}{2} = 2$$

$$d(A_c, D) = \frac{d(A, D) + d(A_c, D) - d(A, A_c)}{2}$$

$$= \frac{5 + 7 - 4}{2} = 4$$

$d(i$

Iteration n-2:

$$\gamma_{Ac} = 2+4 = 6.$$

$$\gamma_B = 2+4 = 6$$

$$\gamma_D = 4+4 = 8$$

m-table:

$$m(Ac, B) = d(Ac, B) - \frac{\gamma_{Ac} + \gamma_B}{n-2},$$

$$= 2 - \frac{6+6}{3-2}$$

$$= 2 - \frac{12}{1} = -10.$$

$$m(Ac, D) = d(Ac, D) - \frac{\gamma_{Ac} + \gamma_D}{n-2},$$

$$= 4 - \frac{6+8}{3-2}$$

$$= 4 - \frac{14}{1} = -10.$$

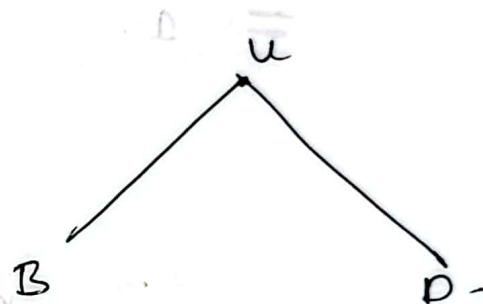
$$m(B, D) = d(B, D) - \frac{\gamma_B + \gamma_D}{n-2},$$

$$= 4 - \frac{6+8}{3-2}$$

$$= \underline{-10.}$$

$M(i,j)$	Ae	B	D
Ae	-	-10	-10
B		-	-10
D			-

B & D are chosen



branch length from u to B is:

$$b(u, B) = \frac{d(B, D)}{2} + \frac{\gamma_B - \gamma_D}{2n - 4}$$

$$= \frac{4}{2} + \frac{6 - 8}{6 - 4}$$

$$= 2 + \left(\frac{-2}{2} \right)$$

$$= 1$$

$$b(u, D) = d(B, D) - d(u, B)$$

$$= 4 - 1 = 3$$

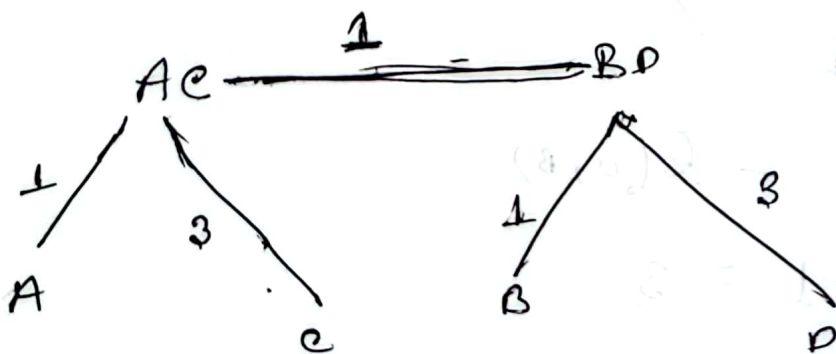
Update the distance matrix:

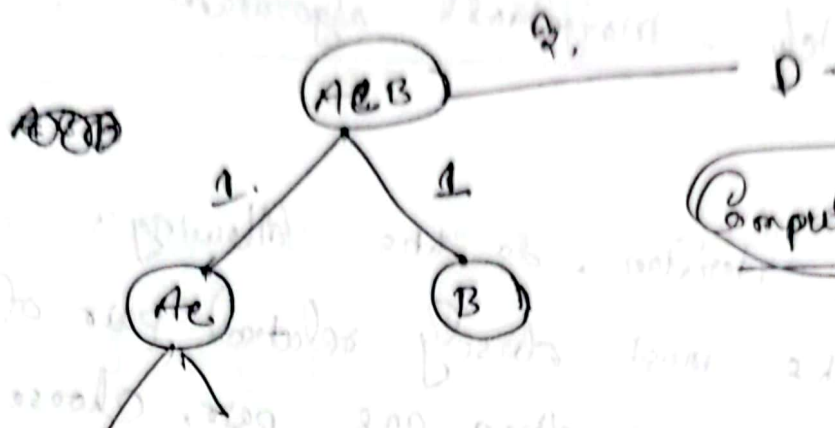
$$d(u, v) = \frac{d(u, i) + d(j, v) - d(i, j)}{2}$$

$$d(BD, Ae) = \frac{d(B, Ae) + d(D, Ae) - d(B, D)}{2}$$

$$= 1.$$

	Ae	BD
Ae	0	1
BD	1	0





Compute m-table

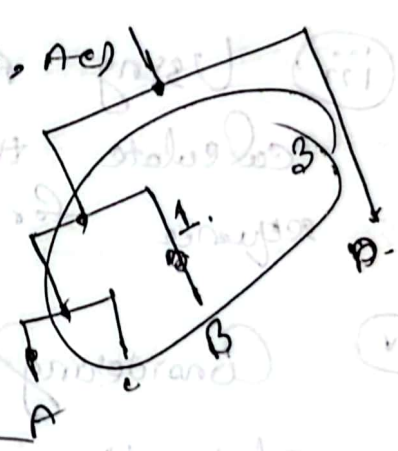
$$b(AEB, Ae) = \frac{d(Ae, B)}{2} + \frac{\delta Ae - \delta B}{2n-4}$$

$$= \frac{2}{2} + \frac{6-6}{2 \cdot 4 - 4}$$

$$= 1$$

$$b(AEB, B) = d(Ae, B) - b(AEB, Ae)$$

$$= 2 - 1 = 1$$



	AEB	D
AEB	-	3
D	3	-

$$d(AEB, D) = \frac{d(Ae, D) + d(B, D) - d(Ae, B)}{2}$$

$$= \frac{4 + 4 - 2}{2} = 3$$

Fitch - Margolash algorithm

① ~~Find~~

For every iteration, do the following:

① Find the most closely related pair of clusters. If there are more than one pair, choose only one pair (A, B).

② Treat the ~~other~~ rest of the clusters (sequences) as W. (one composite cluster). Now calculate the average distance from

1. A to W.

2. B to W.

③ Using A, B & W, draw an unrooted tree, and calculate the branch lengths, using three species/sequence formula,

④ Considering A & B a ~~new~~ composite cluster, (AB), update the evolutionary distance matrix. It means calculate the average distance from AB to all other sequences / clusters.

⑤ Repeat step ① to ④ until there are only three clusters left.

Fitch - Margolish method

① Input: Given a distance evolutionary distance matrix of sequences (cluster single cluster)...

Output: It produces an unrooted phylogenetic tree.

Fitch - Margolish method for three sequences:

① Draw an unrooted tree with three branches emanating from a common node and label ends of branches. Given the closer distance b/w A & B, the branch lengths b/w these sequences are expected to be shorter, as indicated:

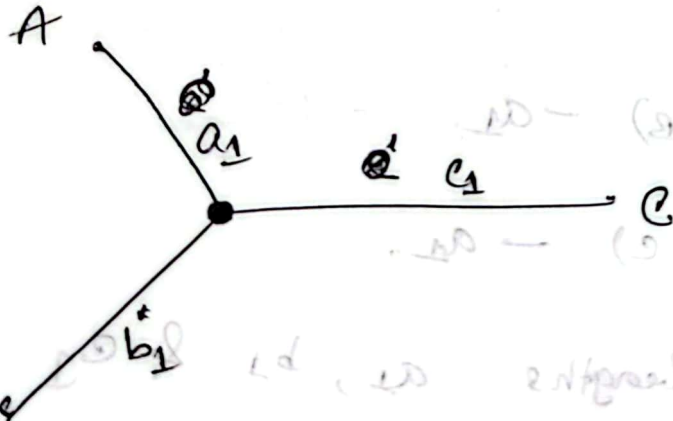


Fig. The Tree showing evolution among A, B and C.

② Branch lengths are calculated as follows;

$$\cancel{d(A, B) = a + b'}$$

$$\cancel{d(A, C) = a + c'}$$

$$d(A, B) = a_1 + b_1 \quad \text{--- (i)}$$

$$d(A, C) = a_1 + c_1 \quad \text{--- (ii)}$$

$$d(B, C) = b_1 + c_1 \quad \text{--- (iii)}$$

$$\textcircled{i} + \textcircled{ii} - \textcircled{iii}$$

$$2a_1 = d(A, B) + d(A, C) - d(B, C)$$

$$\Rightarrow a_1 = \frac{d(A, B) + d(A, C) - d(B, C)}{2}$$

b_1 and c_1 can be calculated similarly or

we can use below formulae:

$$b_1 = d(A, B) - a_1$$

$$c_1 = d(A, C) - a_1$$

If the branch lengths a_1 , b_1 & c_1 are the not the same, it means the species are diverging at different rate of evolution

Distance b/w X and Y are calculated as:

$$d(X, Y) = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d(i, j).$$

Given an evolutionary distance matrix as:

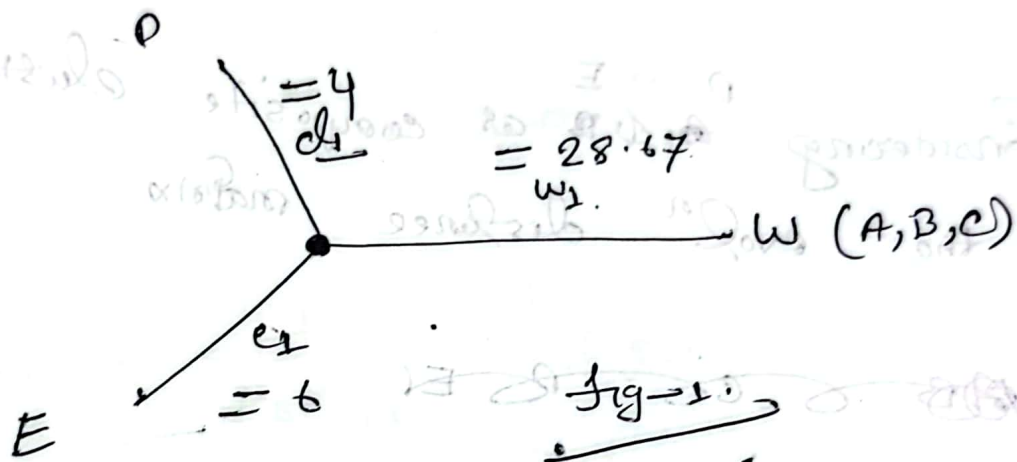
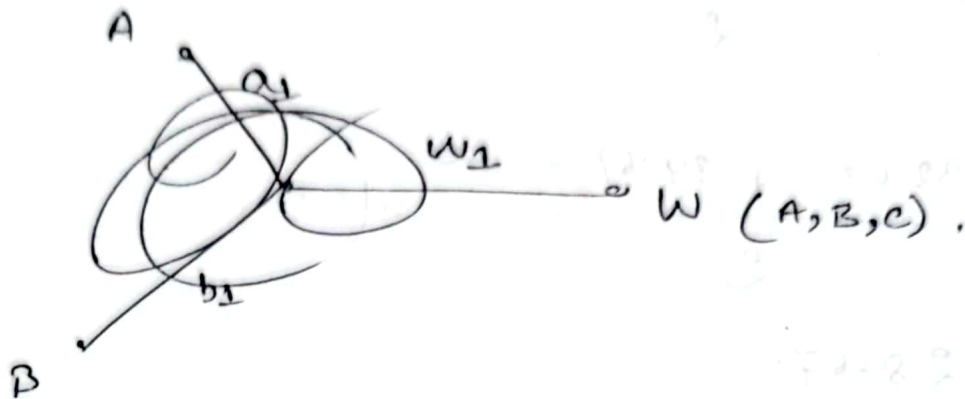
	A	B	C	D	E
A	-	22	39	39	41
B		-	41	41	43
C			-	18	20
D				-	10
E					-

Iteration -1:

Step -1: Find most closely related pairs
is D and E.

$$\text{let } w = \{A, B, C\}.$$

Step-3: D-Tree Drawing & branch length calculation.



$$d_1 = \frac{d(D, E) + d(D, W) - d(E, W)}{2}$$

$$= \frac{10 + 32.67 - 34.67}{2}$$

$$= \frac{8}{2} = 4.$$

$$e_1 = d(D, E) - d_1 = 10 - 4 = 6$$

$$w_1 = d(D, W) - d_1 = 32.67 - 4 = 28.67$$

$$w_3 = \frac{d(D, W) + d(E, W) - d(D, E)}{2}$$

$$= \frac{32.67 + 34.67 - 10}{2}$$

$$= 28.67$$

Step-4: Considering ^D & ^E as composite cluster ^{DE},
update the evolⁿ distance matrix.

~~DE~~ ~~C~~ ~~B~~ ~~A~~

	A	B	C	D, E
A	-	22	39	40
B		-	41	42
C			-	49
D, E				-

$$Q(A, DE) = \frac{1}{N_A N_{DE}} \sum_{i \in A, j \in DE} d(i, j)$$

$$= \frac{1}{1 \times 2} [d(A, D) + d(A, E)]$$

$$= \frac{1}{2} (39 + 41)$$

$$= \frac{1}{2} \times 80 = 40$$

$$Q(B, DE) = \frac{1}{N_B N_{DE}} [d(B, D) + d(B, E)]$$

$$= \frac{1}{1 \times 2} [41 + 43]$$

$$= \frac{1}{2} \times 84 = 42$$

$$Q(C, DE) = \frac{1}{N_C N_{DE}} [d(C, D) + d(C, E)]$$

$$= \frac{1}{1 \times 2} [18 + 20]$$

$$= \frac{1}{2} \times 38 = 19$$

~~Final~~
Iteration - 2:

Step - 1: most closely related pair (C, DE)
1st, $w = \{A, B\}$

Step - 2:

$$Q(C, w) = \frac{1}{N_C N_w} [d(C, A) + d(C, B)]$$

$$= \frac{1}{1 \times 2} [39 + 42]$$

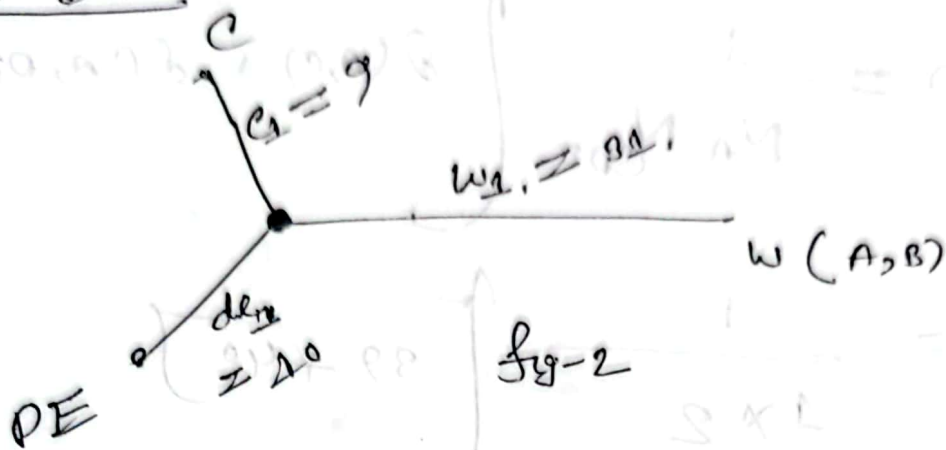
$$= 40.5$$

$$Q(DE, w) = \frac{1}{N_{DE} N_w} [d(DE, A) + d(DE, B)]$$

$$= \frac{1}{1 \times 2} [40 + 42]$$

$$= 41$$

Step-3:



$$c_1 = \frac{d(c, w) + d(c, DE) - d(DE, w)}{2}$$

$$= \frac{40 + 19 - 41}{2}$$

$$d_1 = d(c, DE) - c_1 = 19 - 9 = 10$$

$$w_1 = d(c, w) - c_1 = 40 - 9 = 31$$

Step-4:

Consider ~~c & DE~~ as a make c & DE as a composite cluster and update the distance matrix

	A	B	(c, DE)
A	—	22	39.5
B		—	41.5
(c, DE)			—

$$D(A, CDE) = \frac{1}{N_A N_{CDE}} \left[D(A, C) + D(A, DE) \right]$$

$$= \frac{1}{1 \times 2} \left[39 + 40 \right]$$

$$= \frac{1}{2} \times 79 = 39.5$$

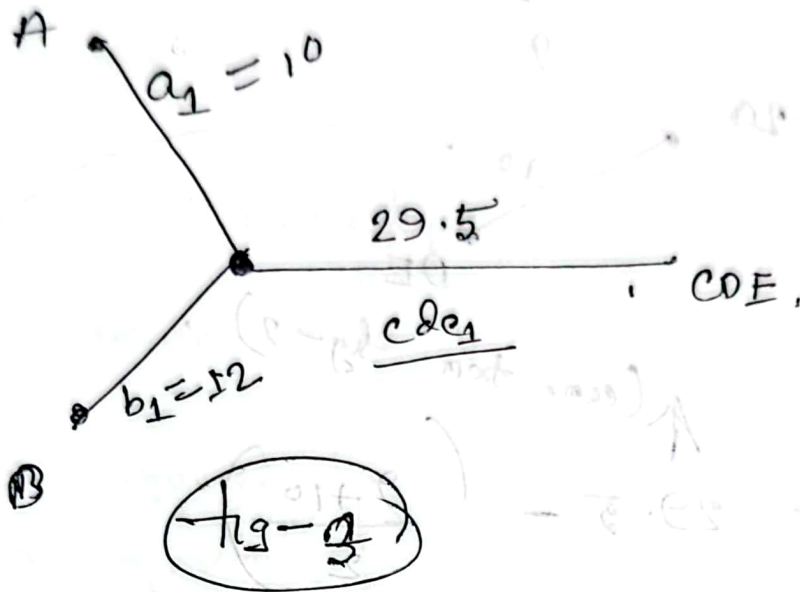
$$D(B, CDE) = \frac{1}{N_B N_{CDE}} \left[\cancel{D(B, CDE)} + \cancel{D(B, C)} + D(B, C) + D(B, DE) \right]$$

$$= \frac{1}{1 \times 2} \left[41 + 42 \right]$$

$$= \frac{83}{2} = 41.5$$

Horizon-3:

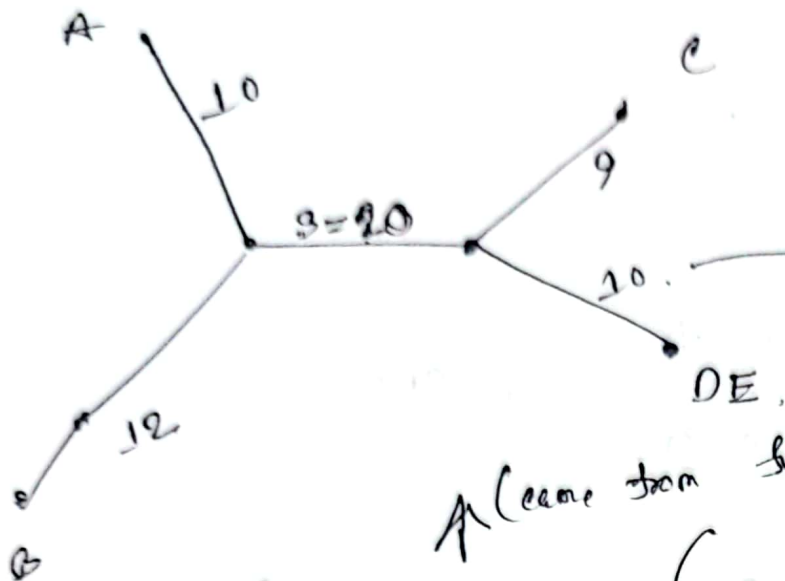
Step-1:



$$a_1 = \frac{d(A, CDE) + d(A, B) - d(B, CDE)}{2}$$
$$= \frac{39.5 + 22 - 41.5}{2}$$
$$= 10$$

$$b_1 = d(A, B) - a_1 = 22 - 10 = 12$$

$$cde_1 = d(A, CDE) - a_1 = 39.5 - 10 = 29.5$$

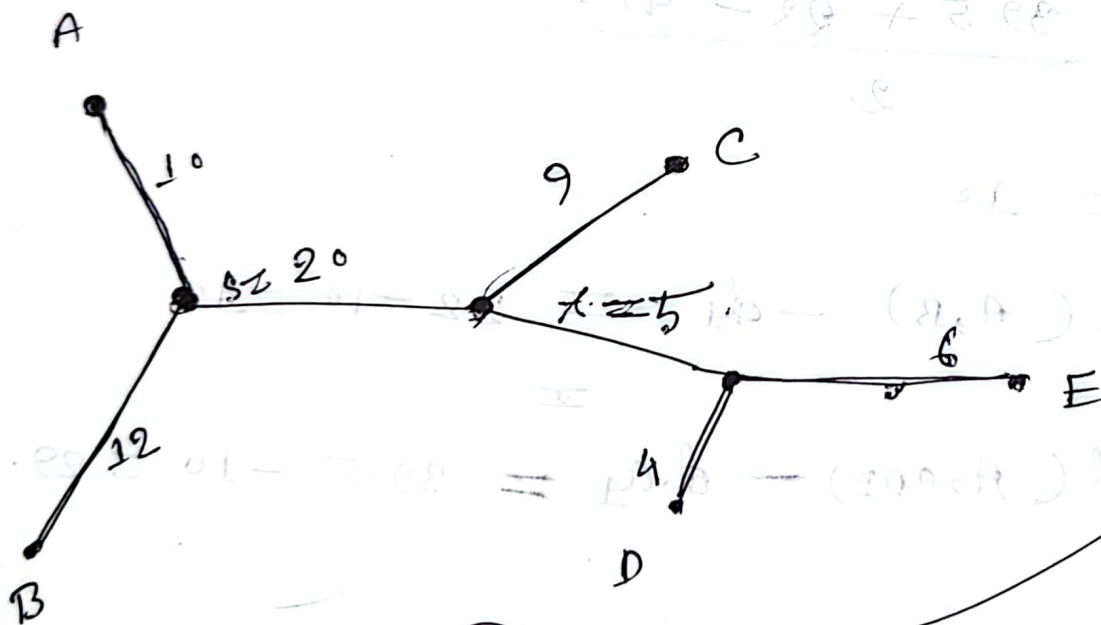


↑ (can't join $9-10$)

$$S = 29.5 - \left(\frac{9+10}{2} \right)$$

$$= 29.5 - \frac{19}{2}$$

$$= 29.5 - 9.5 = 20$$



$$S = 49 - \left(\frac{4+6}{2} \right)$$

$$= 49 - 5 = 44$$

Sequence: An arrangement of two or more related things in a successive order.

A Sequence is basically an array of nucleotide bases or amino acids.

DNA sequence	}	Protein sequence
RNA sequence		

Sequence alignment:

The alignment of biological sequences is probably the most accomplished and most important in the field of bioinformatics. Their ultimate goal is to determine the ^{identity} similarity b/w different sequences.

Sequence similarity Vs. Sequence identity:

Both words have same meaning in nucleotide sequences.

Two DNA sequences can share a high degree of sequence identity / sequence similarity, but GP means the same thing.

For protein sequence, both words have different meanings.

Sequence identity: % of aligned residues that are identical.

Sequence similarity: % of aligned residues that are similar in characteristics.

Q Given three sequences $S_1, S_2 \& S_3$ ~~A, B & C~~ as:

S_1 : A A G G C T T

S_2 : A A G G C

S_3 : A A G G C A T

GP $S_1 = S_2 \& S_2 = S_3$

then $S_1 = S_3$? (In terms of identity distance measure).

$$\text{identity}(s_1, s_2) = 100\%$$

$$= \frac{5 \text{ Identical nucleotides}}{\min(\text{len}(s_1), \text{len}(s_2))}$$

$$\text{identity}(s_2, s_3) = 100\%$$

$$\text{identity}(s_1, s_3) = \frac{6 \text{ Identical nucleotides}}{\min(\text{len}(s_1), \text{len}(s_3))} \times 100\%$$

$$= \frac{6}{7} \times 100\%$$

$$= 85.7\%$$

Hence, $s_1 \neq s_3$ in terms of identity.

$\therefore$ 100% identity does not mean that two sequences are the same.

Terms

Biological event

match

conservation

mismatch

substitution

Gap

Insertion / Deletion

Gap penalty:

Cost to introduce gap

constant

Linear gap

Affine gap penalty

Linear gap penalty of whole

sequence $\geq$ no. of gap $\times$ constant

↓
gap cost for each gap.

Affine gap penalty of whole sequence

$\geq N \times \text{gap opening penalty} + E \times \text{gap extension penalty}$

N = number of opening gaps.

E = number of extension gaps.

Q1:

A G G C T A C T _ T _ T C A

G G C T A C T A T A T C A

Affine gap penalty

$$= N \times \text{gap opening penalty} + E \times \text{gap extension penalty}$$

Here, $N = 2$ and $E = 0$

$$\text{gap opening penalty} = -5$$

$$= 2 \times (-5) + 0 \times \text{gap extension penalty}$$

$$= -10$$

extension penalty

Q2:

A G G C T A C T T T _ _ C A

G G C T A C T A T A T C A

Affine gap penalty

$$N = 1$$

$$E = 1$$

$$= N \times \text{gap opening penalty} + E \times \text{gap extension penalty}$$

$$= 1 \times (-5) + 1 \times (-1)$$

$$= -5 - 1$$

$$= -6$$

$$\underline{Q} \quad S_1 = PRT _ _ _ EINS$$

$$S_2 = PRT PWSEIN _$$

$$\text{gap penalty } (S_1) = N \times \text{gap opening penalty} +$$

$$N = 1$$

$$E = 2$$

$$\text{gap opening penalty} = -11$$

$$\text{gap extension penalty} = -1$$

$$= 1 \times (-11) + 2 \times (-1)$$

$$= -13$$

$$\text{gap penalty } (S_2) = N \times \text{gap opening penalty} + E \times \text{gap extension penalty}$$

$$N = 1$$

$$E = 0$$

$$= 1 \times (-11) + 0$$

$$= -11$$

Total

$$\text{gap penalty} = \text{gap penalty } (S_1) + \text{gap penalty } (S_2)$$

$$= (-13) + (-11)$$

$$= -24$$

P = Proline (Pro).

R = Arg. (Arginine)

T = Thr (Threonine).

E = glu (glutamate).

I = Ile (Isoleucine).

N = Asn (Asparagine).

Matching

$$\text{Score} = 7 + 5 + 5 + 5 + 4 + 6 = 32$$

$$\text{Score} = 32 - 24 = 8$$

Scoring matrix.

we perform sequence alignment.

~~we perform
seq alignment~~

0.841836

0.841

$$\begin{array}{c} 0 \\ \hline 11 \\ \hline 30 \end{array} \times \begin{array}{c} E \\ \hline 98 \\ \hline 225 \end{array}$$